

FITBUDDY

AI Fitness Plan Generator using Gemini Models

Final Project Report

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1. INTRODUCTION

1.1 Project Overview

FitBuddy – AI Fitness Plan Generator using Gemini Models is a web-based application that uses Generative AI to create personalized fitness plans. The system collects user information such as name, age, weight, fitness goal, and preferred workout intensity and generates a personalized 7-day workout plan. The application also provides nutrition and recovery guidance and allows users to provide feedback so that the workout plan can be updated according to their requirements.

1.2 Purpose

The purpose of FitBuddy is to simplify fitness planning by providing personalized workout guidance through Gemini AI. The application aims to reduce the difficulty of creating suitable workout routines and provide users with easy-to-follow fitness, nutrition, and recovery guidance.

2. IDEATION PHASE

2.1 Problem Statement

Users often find it difficult to create workout routines that match their individual fitness goals, preferences, and workout intensity. Generic fitness plans may not adapt easily to changing requirements or user feedback.

2.2 Empathy Map Canvas

The FitBuddy empathy map identifies the target user's thoughts and feelings, what they hear and see, what they say and do, their pain points, and the gains expected from a personalized fitness solution.

- **Think & Feel:** Wants to achieve fitness goals, wants a personalized workout plan, feels unsure about suitable exercises, and wants clear guidance.
- **Hear:** Friends, fitness influencers, and other sources provide different workout and diet advice.
- **See:** Many workout plans, fitness apps, exercise routines, and nutrition suggestions are available online.
- **Say & Do:** Searches for workouts, tries different exercises, adjusts intensity, and looks for nutrition guidance.
- **Pain:** Difficulty creating a personalized plan, confusion about intensity, uncertainty about nutrition/recovery, and difficulty maintaining consistency.
- **Gain:** Personalized 7-day workout plan, clear guidance, nutrition and recovery tips, and feedback-based plan updates.

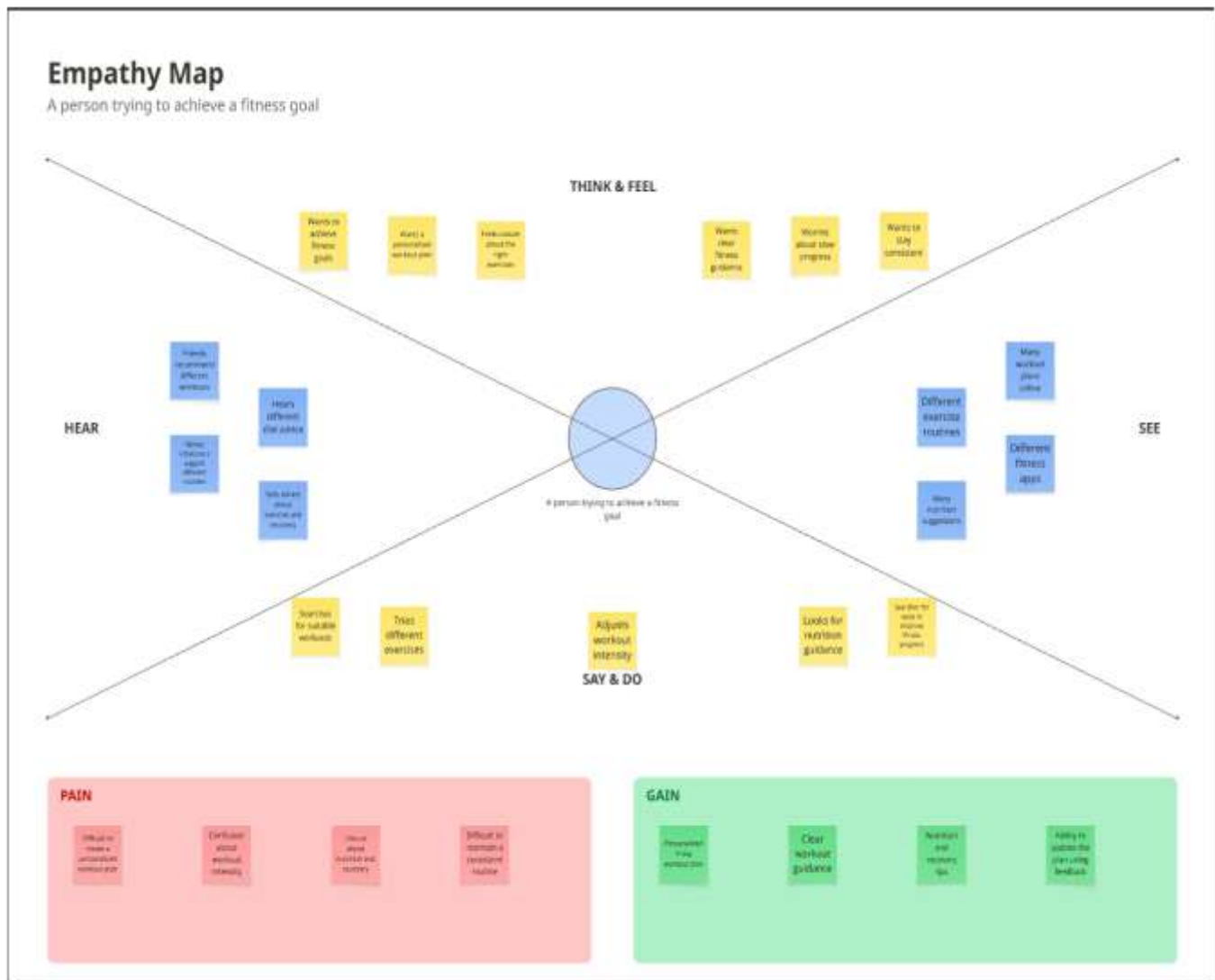


Figure 2.2: FitBuddy Empathy Map Canvas

2.3 Brainstorming

The team brainstormed fitness-related features and grouped them into AI Fitness Planning, Nutrition & Recovery, Progress & Personalization, User Engagement, and Administration. The ideas were then prioritized based on importance and feasibility.

- AI Fitness Planning: AI workout plan, 7-day fitness plan, goal-based workouts, intensity selection, personalized exercise suggestions.
- Nutrition & Recovery: Nutrition tips, recovery guidance, rest-day recommendations.
- Progress & Personalization: Fitness progress tracking, workout history, saved workout plans.
- User Engagement: Fitness chatbot, workout reminders.
- Administration: Admin dashboard, user management.

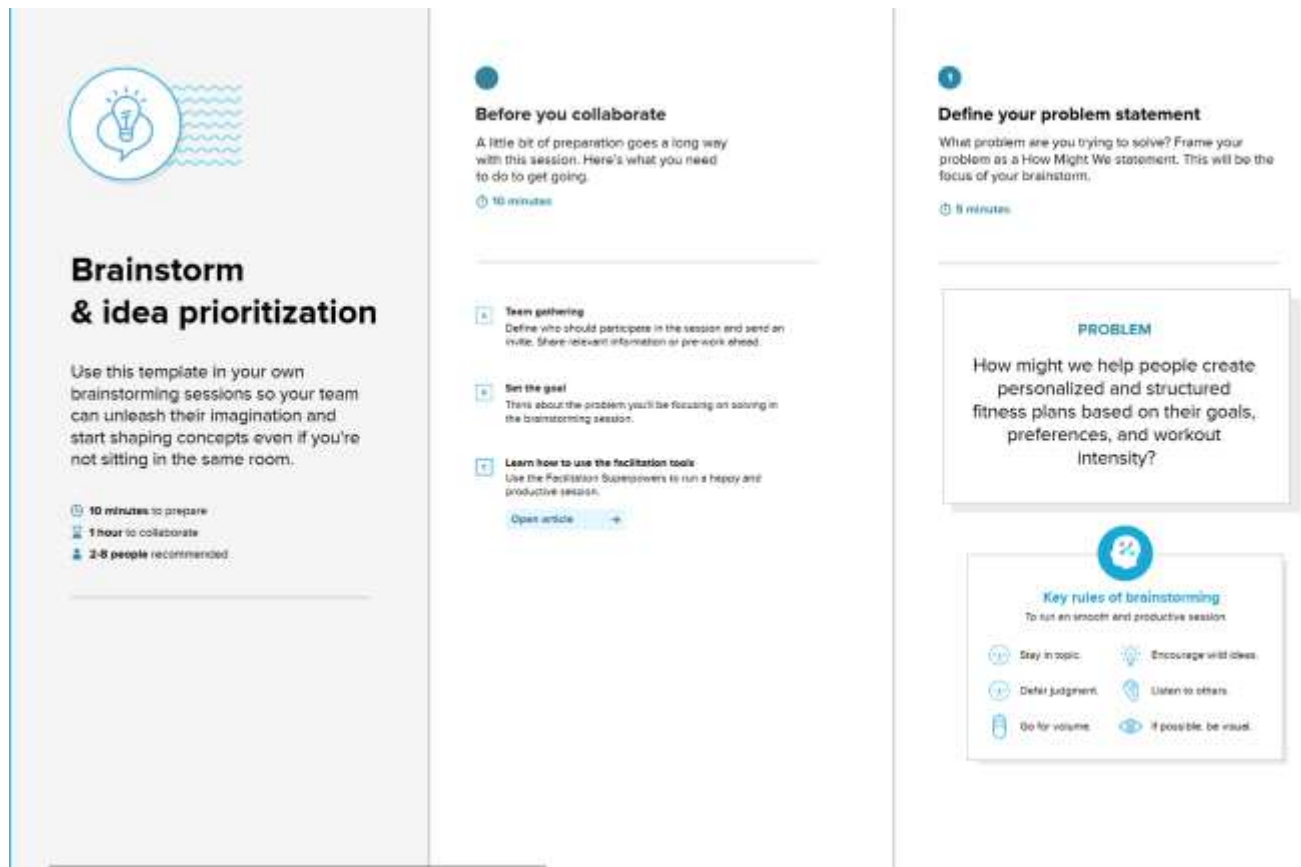


Figure 2.3: FitBuddy Brainstorming and Idea Prioritization

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The final report format requires a Customer Journey Map. A completed Customer Journey Map was not included in the source template material provided for this report, so this section is reserved for the completed FitBuddy journey-map artifact.

Stage	User Action	User Need	Pain Point	FitBuddy Opportunity
Awareness	Looks for fitness guidance	Simple guidance	Too many generic sources	Present a simple AI fitness solution
Input	Enters fitness information	Personalization	Unsure what details are needed	Collect age, weight, goal and intensity
Generation	Requests a workout plan	Suitable routine	Generic plans may not fit	Generate a personalized 7-day plan
Feedback	Provides plan feedback	Adjustments	Difficult to modify routines	Regenerate the plan using feedback

Follow-up	Views guidance/history	Consistency	Hard to remember previous plans	Store and retrieve workout information
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3.2 Solution Requirement

Functional Requirements

FR No.	Functional Requirement	Sub-Requirements
FR-1	User Profile & Fitness Input	Enter name, age, weight; select fitness goal; select workout intensity; submit fitness information.
FR-2	AI Workout Plan Generation	Generate personalized 7-day plan; goal-based exercises; adjust exercises to intensity; use Gemini AI.
FR-3	Nutrition & Recovery Guidance	Provide nutrition tips, recovery guidance, and rest-day recommendations.
FR-4	Workout Plan Feedback	Accept user feedback; modify the plan; regenerate an updated plan using Gemini AI.
FR-5	Workout Plan History	Store generated plans; view previous plans; retrieve user workout information.
FR-6	Admin Management	View registered users; view generated workout plans; manage stored information.

Non-Functional Requirements

NFR	Description
NFR-1 Usability	Simple and user-friendly interface.
NFR-2 Security	Protect user information, validate inputs, and protect API credentials.
NFR-3 Reliability	Handle invalid/incomplete inputs and provide consistent processing.
NFR-4 Performance	Provide efficient application and API responses.
NFR-5 Availability	Available whenever the application server is running.
NFR-6 Scalability	Support additional users and future features.

3.3 Data Flow Diagram

The FitBuddy DFD represents the flow between the user, web application, FastAPI backend, Gemini AI, and SQLite database. Inputs include name, age, weight, fitness goal, intensity, and feedback. Outputs include a 7-day plan, nutrition and recovery recommendations, and updated plans.

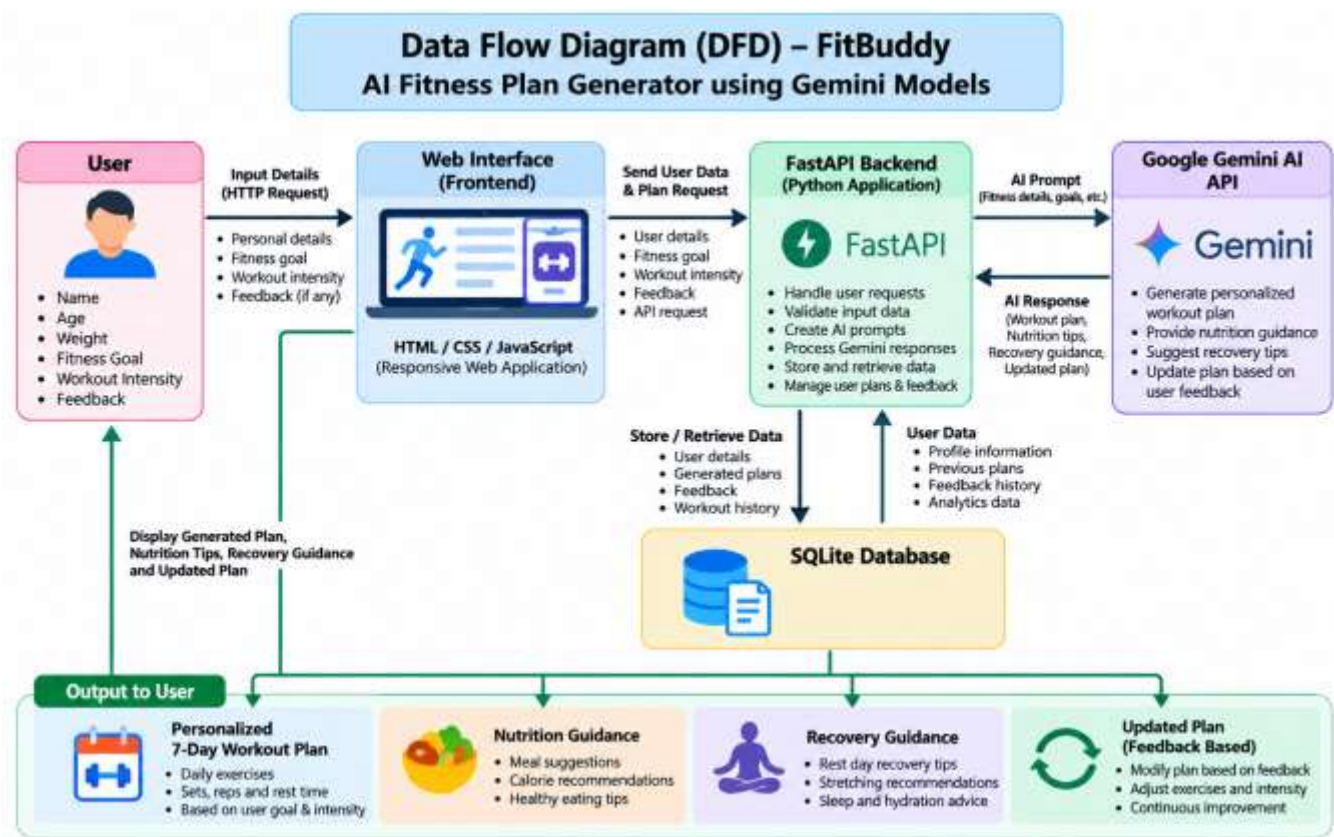


Figure 3.3: FitBuddy Data Flow Diagram

3.4 Technology Stack

Component	Technology
Frontend	HTML5, CSS3, JavaScript
Backend	Python, FastAPI
AI	Google Gemini AI
Database	SQLite
ORM / Database Access	SQLAlchemy
AI API	Google Gemini API
Application Architecture	Web-based client-server architecture
Dynamic Templates	Jinja2
Server	Uvicorn / FastAPI server

4. PROJECT DESIGN

4.1 Problem Solution Fit

FitBuddy focuses on understanding the target user's fitness goals, constraints, behaviors, triggers, available alternatives, and pain points before defining the solution. The solution is designed to provide personalized fitness planning that fits the user's needs.



Figure 4.1: FitBuddy Problem-Solution Fit Canvas

4.2 Proposed Solution

FitBuddy is an AI-powered web application that uses Gemini AI to generate a personalized 7-day workout plan based on the user's name, age, weight, fitness goal, and preferred workout intensity. Users can provide feedback to update their plan and receive nutrition and recovery guidance.

- Personalized 7-day workout generation.
- Goal-based and intensity-based exercise recommendations.
- Nutrition and recovery guidance.
- Feedback-based workout plan modification.
- Storage and retrieval of workout information.

4.3 Solution Architecture

The FitBuddy architecture connects the user interface, FastAPI backend, Gemini AI API, and SQLite database. User fitness information and feedback are sent to the backend, which prepares AI prompts, processes Gemini responses, and stores or retrieves relevant user and plan information.

- User → Web Interface: Name, age, weight, fitness goal, intensity, and feedback.
- Web Interface → FastAPI: Fitness details and workout-plan requests.

- FastAPI → Gemini AI: Formatted AI prompt.
- Gemini AI → FastAPI: Workout plan, nutrition guidance, recovery guidance, and updated plan.
- FastAPI ↔ SQLite: User details, generated plans, feedback, and workout history.

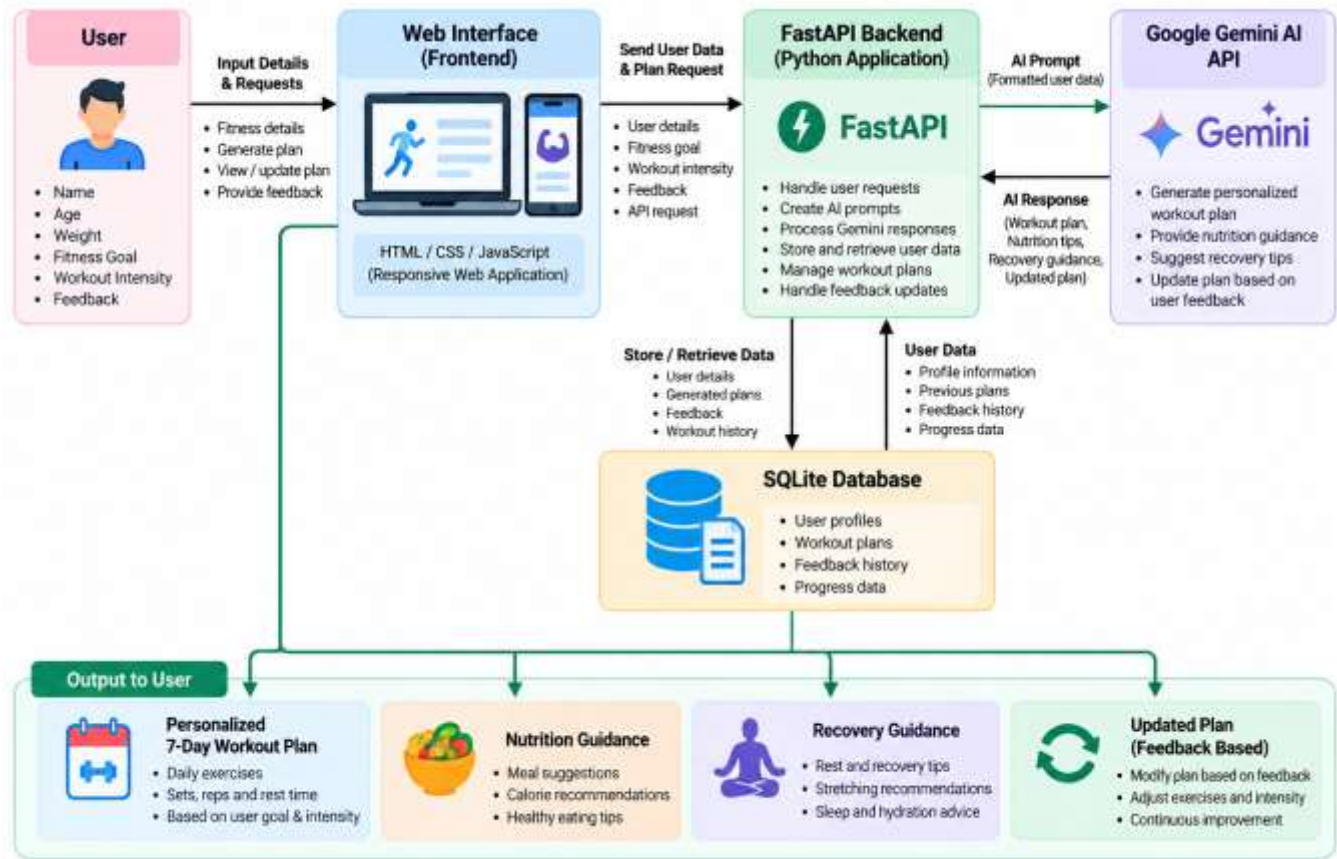


Figure 4.3: FitBuddy Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was planned across two sprints using Agile/Scrum concepts.

Product Backlog and Sprint Planning

Sprint	Epic	User Story	Story Points	Priority
Sprint-1	User Profile & Fitness Input	USN-1: Enter name, age, weight, goal and intensity.	2	High
Sprint-1	AI Workout Plan Generation	USN-2: Generate personalized 7-day workout plan.	5	High
Sprint-1	Nutrition & Recovery	USN-3: Receive nutrition and recovery guidance.	3	High

Sprint-1	Workout Plan Feedback	USN-4: Update plan based on user feedback.	3	High
Sprint-2	Workout History	USN-5: View previously generated workout plans.	3	Medium
Sprint-2	Admin User Management	USN-6: Admin can view registered users.	2	Medium
Sprint-2	Admin Plan Management	USN-7: Admin can view generated plans.	2	Medium

Sprint Tracker

Sprint	Total Points	Duration	Start Date	End Date	Completed	Release Date
Sprint-1	13	6 Days	05 Sep 2026	10 Sep 2026	13	10 Sep 2026
Sprint-2	7	6 Days	12 Sep 2026	17 Sep 2026	7	17 Sep 2026

Velocity

Total Story Points = 20

Number of Sprints = 2

Velocity = $20 / 2 = 10$ Story Points per Sprint

Burndown Chart

A burndown chart represents the work remaining to be completed over time. For FitBuddy, the remaining work is 20 points at project start, 7 points after Sprint-1, and 0 points after Sprint-2.

Project Stage	Story Points Remaining
Project Start	20
After Sprint-1	7
After Sprint-2	0

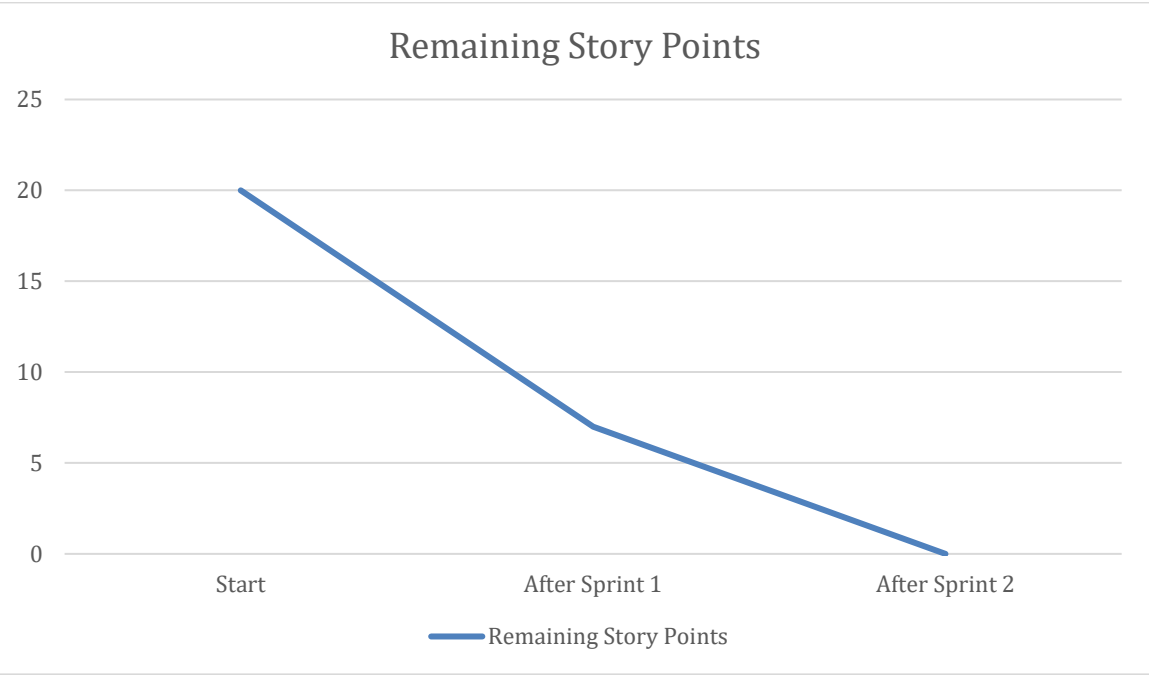


Figure 5.1: FitBuddy Burndown Chart

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Scenario	Test Steps	Expected Result	Actual Result	Result
FT-01	Fitness Text Input Validation	Enter valid/invalid name values.	Valid text accepted; invalid input handled.	Valid name accepted and invalid/empty input handled.	PASS
FT-02	Fitness Number Input Validation	Enter valid/invalid age and weight.	Valid values accepted; invalid values rejected.	Valid age and weight accepted; invalid values rejected.	PASS
FT-03	Goal & Intensity Selection	Select goal and intensity.	Selections processed correctly.	Goal and intensity processed successfully.	PASS
FT-04	Workout Plan Generation	Enter complete details and generate.	Personalized 7-day plan generated.	Personalized 7-day plan generated successfully.	PASS
FT-05	Feedback-Based Plan Update	Submit feedback such as more cardio.	Plan updated according to feedback.	Plan updated successfully based on feedback.	PASS

FT-06	Gemini API Connection	Submit a valid request.	Gemini API responds successfully.	Gemini API responded successfully.	PASS
PT-01	Workout Plan Response Time	Measure plan generation time.	Acceptable response time.	Plan generated within an acceptable response time.	PASS
PT-02	API Request Performance	Send multiple requests.	No unexpected errors or major slowdown.	Multiple requests processed successfully.	PASS
PT-03	Database Operation	Generate plan and check SQLite.	Data stored and retrieved correctly.	User and plan information stored and retrieved successfully.	PASS

7. RESULTS

7.1 Output Screenshots

This section should contain screenshots captured from the working FitBuddy application. The following placeholders identify the recommended screenshots to insert.

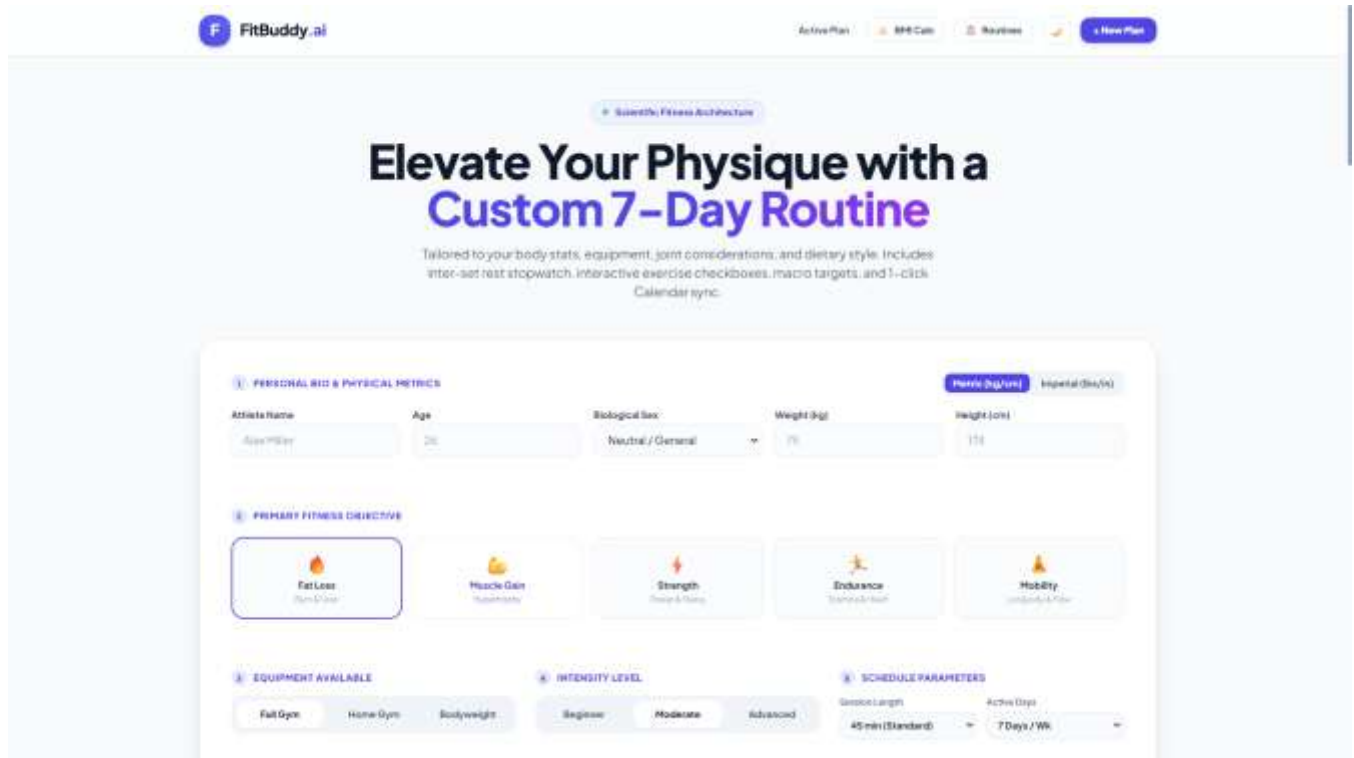


Figure 7.1: FitBuddy Home / Landing Page.

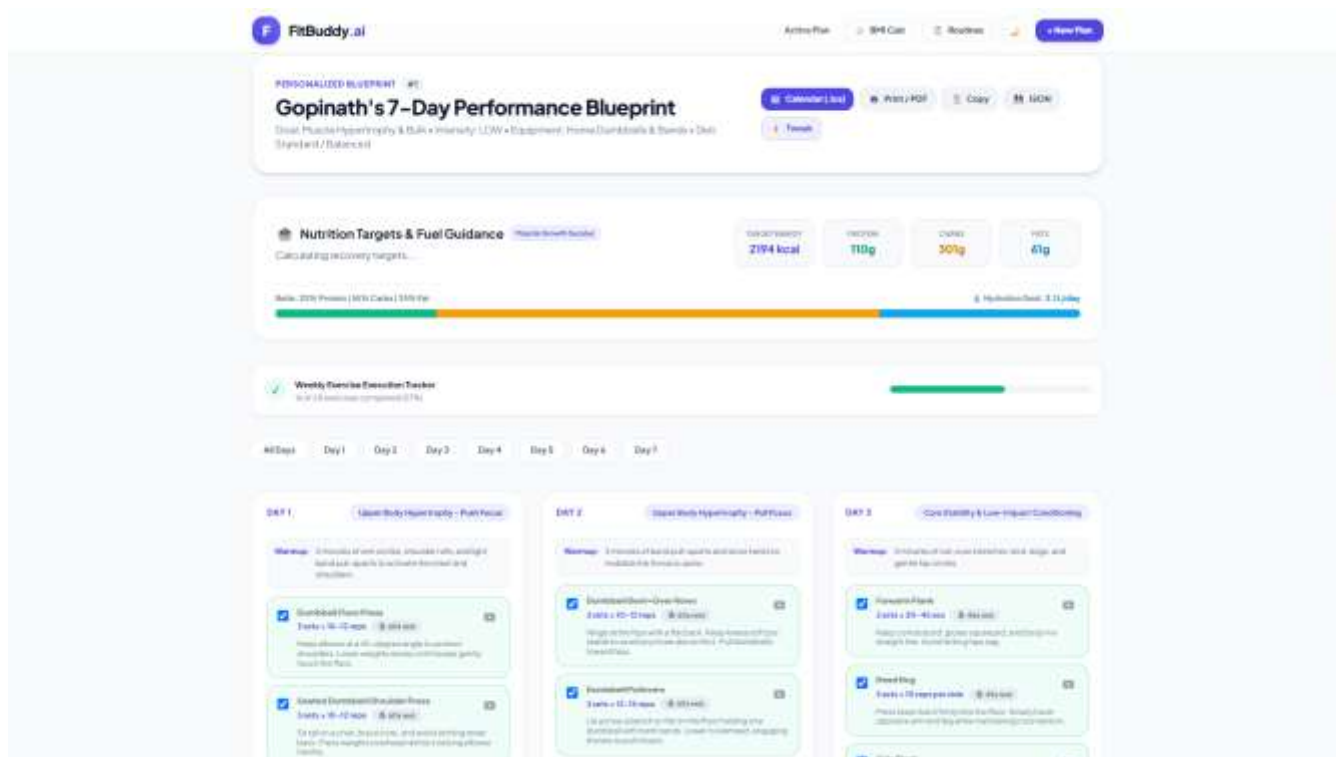


Figure 7.4: Generated 7-Day Workout Plan.

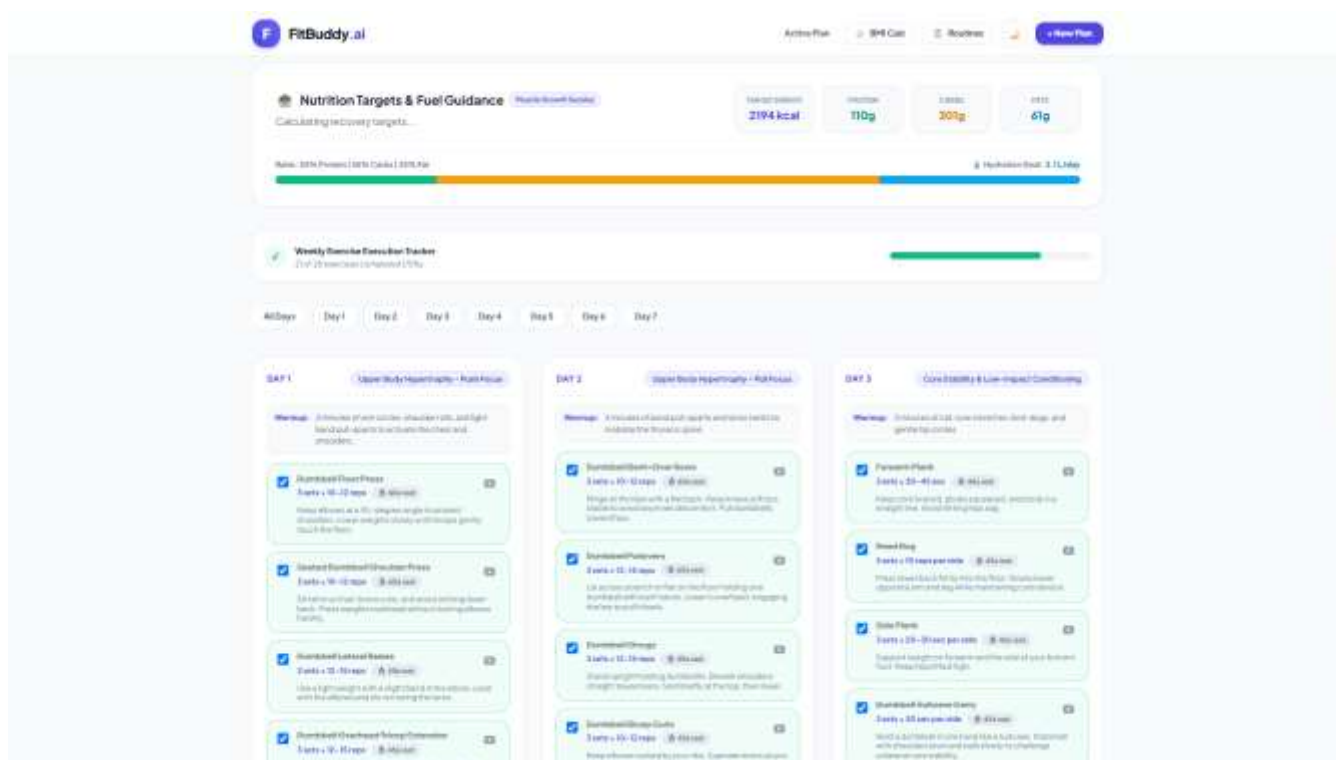


Figure 7.5: Nutrition Guidance.

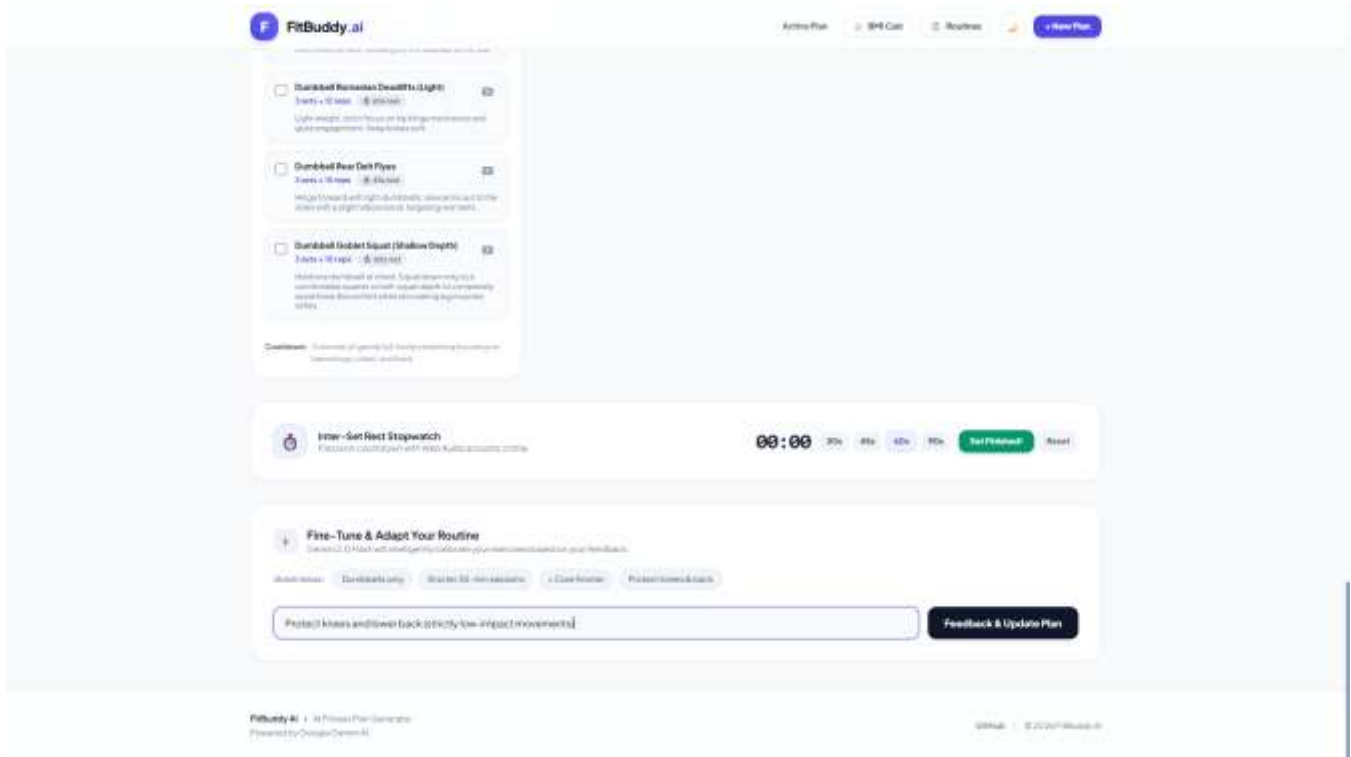


Figure 7.6: Feedback and Updated Workout Plan.

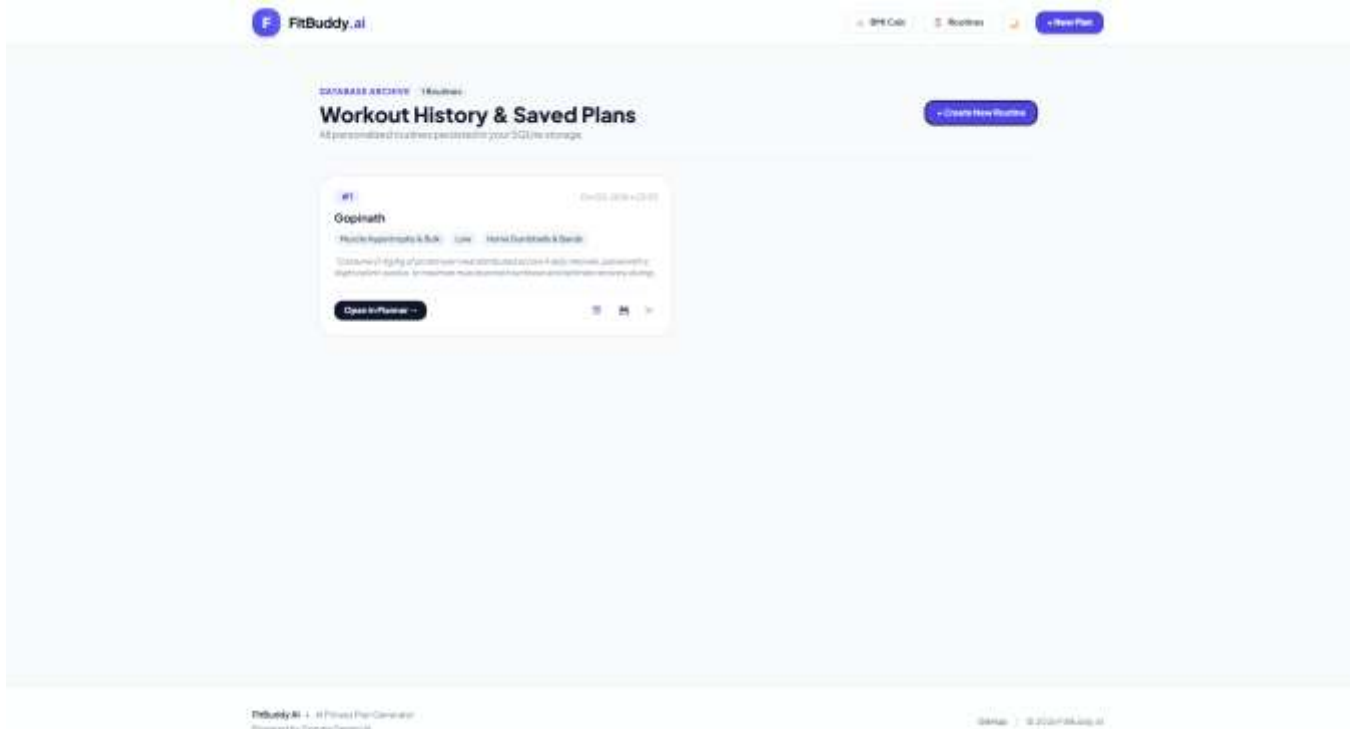


Figure 7.7: Workout History.

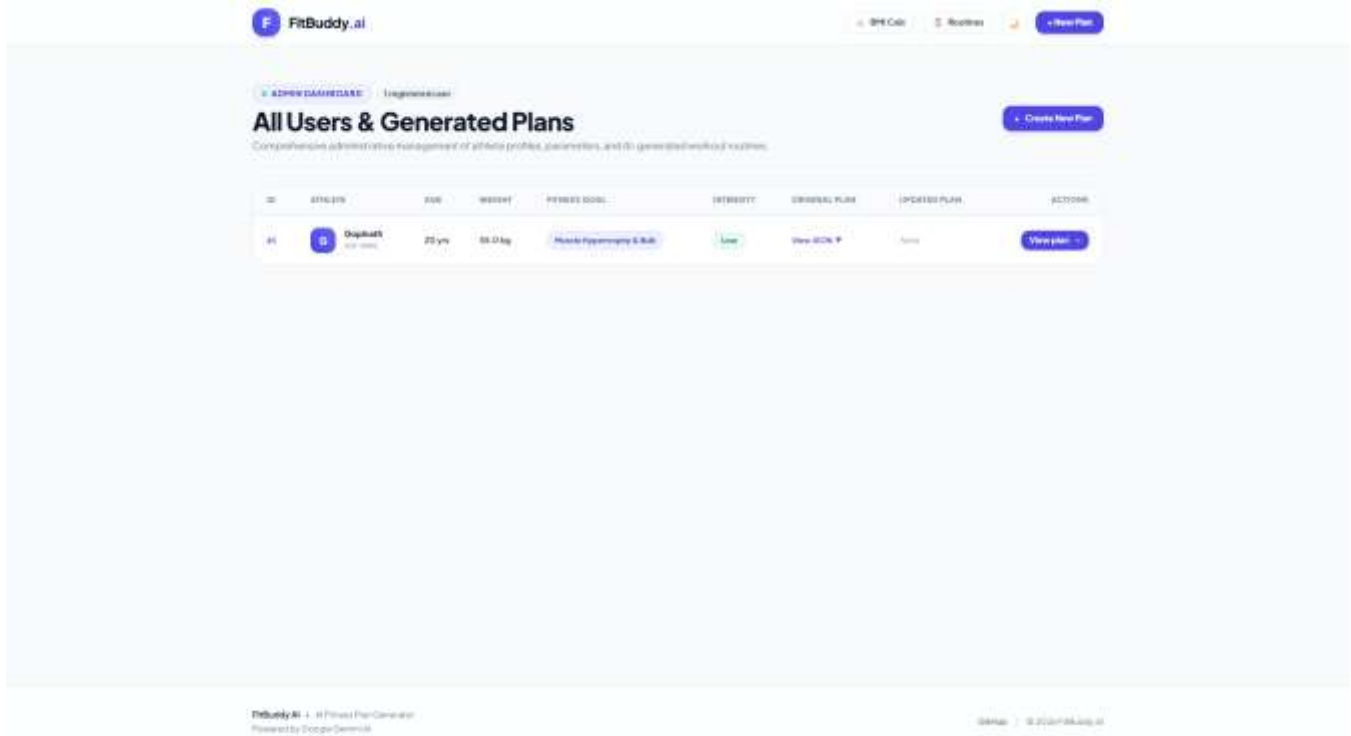


Figure 7.8: Admin Dashboard.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Provides personalized workout plans.
- Uses Gemini AI for dynamic plan generation.
- Supports different fitness goals and workout intensities.
- Provides nutrition and recovery guidance.
- Allows feedback-based workout-plan updates.
- Provides a simple and interactive web interface.
- Stores workout information using SQLite.

Disadvantages

- Requires an internet connection for Gemini AI requests.
- AI-generated recommendations may require user verification.
- Response time can depend on the Gemini API.
- The current prototype has limited advanced progress-tracking features.
- The application is intended for general wellness guidance and not medical advice.

9. CONCLUSION

FitBuddy – AI Fitness Plan Generator using Gemini Models provides a simple AI-based approach to personalized fitness planning. The application generates 7-day workout plans based on user information and fitness preferences while also providing nutrition and recovery guidance. The feedback mechanism allows users to modify their plans according to their requirements. The project demonstrates the integration of FastAPI, Gemini AI, HTML/CSS/JavaScript, and SQLite to develop a practical Generative AI web application.

10. FUTURE SCOPE

- Advanced fitness progress tracking.
- Workout reminders and notifications.
- Gamification and achievement badges.
- AI fitness chatbot.
- Dedicated Android/iOS application.
- More detailed user profiles.
- Workout analytics and progress graphs.
- More advanced AI personalization.
- Cloud deployment and scalable database.
- Integration with fitness and wearable devices.

11. APPENDIX

Source Code

GitHub Repository: [GitHub repository link](#)

Dataset Link

Not Applicable / No external dataset used. FitBuddy generates plans dynamically using user-provided fitness information and Gemini AI.

GitHub & Project Demo Link

GitHub & Project Demo : [GitHub Documents Link](#)

Wellness Disclaimer: FitBuddy is intended to provide general fitness and wellness guidance. It is not a substitute for professional medical advice, diagnosis, or treatment.